MFMS Test Record

Testing lead: Tjimezeri Muhupua Part 1: integration and input-validation checks run while the modules were being joined into one program Date: October 2026 Method: manual testing of the compiled program (./mfms), entering valid input and bad input (letters, negatives, ze

A fresh clone of the repository built with no errors and no warnings. ro, empty lines, spaces only, duplicates, wrong menu numbers) in each module.

Build used:

gcc -std=c99 -Wall -Wextra -pedantic -o mfms main.c utils.c employees.c budget.c suppliers.c assets.c reports.c

Introduction and methods

The program was run by executing the compiled version (./mfms) and by providing valid and bad input in every module. Bad input consisted of letters where numbers should be, negative numbers, zero, blank lines, lines that contained only spaces, duplicate IDs, menu numbers that do not exist in the program. The program was compiled with gcc -std=c99 -Wall -Wextra -pedantic -o mfms main.c utils.c employees.c budget.c suppliers.c assets.c reports.c. All code in the repository was pulled and compiled with no errors or warnings.

Main menu

The main menu rejected abc, 0, -1, 99, 99999999 and n, then displayed it again without crashing. Options 1 to 5 each accessed the correct module and then popped back to the main menu. Option 6 asked for confirmation, with a wrong answer (x) asking again, n returning you to the menu, and y quiting with a farewell message.

Employee Management

Rejection of invalid employee; acceptance of duplicate id. Rejection of empty name, acceptance and rejection of name consisting of only spaces at first and then after fix to defect 2. Rejection of basic salary -5000, abc or 0; rejection of negative house allowance; rejection of negative transport allowance. Rejection of ID prompt of -1, abc, 99999999999. Display of all employees with gross salary in employee list. Search by name, department and id, print "Employee not found" if no match found. Rejection of search menu choice 105 and employee menu input abc and 99. Display of gross, tax and net wages for basic 35000 house 4000 and transport 2000 is

41000, 12300 and 28700 respectively, consistent with original. Display of report for total, average, maximum, minimum salary; with one employee all three salary values are 104000.

Budget Management

Initially, typing any character into the alphabet menu made the program go into an endless loop, as in defect 1. Once this was fixed, abc, 99 and 20 were not allowed in the alphabet menu, but did not cause a loop. A valid budget (Finance, 500000) was stored, and an empty department name (i.e. '') was not allowed in the department menu, but a name of (say) ' ' was accepted, and stored unchanged. A department name with a space in it (e.g. 'IT Department') was preserved intact. In the budget menu, an allocated budget of -500, 0, abc or 200000000000 was refused, and in the expenditure menu, an expenditure of -500, 0 or abc was refused. The program after spending 42000 on the 500000 budget showed 458000 left, WITHIN BUDGET. After recording another expenditure of 520000 (i.e. A total expenditure of 562000), it showed 62000 over budget, and displayed the department as over budget in the list of over-budget departments. The total spent and remaining budgets both showed correctly -62000 / 62000 over budget; and a line was not added to the list of over-budget departments when there was no budget. Attacking a department which had not been allocated a budget produced an error message, rather than crashing.

Supplier Management

The initial supplier version accepted a phone number with letters so it was replaced, as described in defect 3. The empty name, phone number with letters, and 10 digit number, and no @, empty email were rejected, a new valid supplier was added, and searching by name returned the correct supplier or "No matching suppliers found.", editing, deleting, and menu input of abc and 99 were all correct and repeated 10,000 times, entering menu was correct and repeated 1,000 times.

Asset Management

A valid asset was entered. All condition entries, letters (good) and a duplicate ID and name (empty) were rejected. All purchase amounts of -500, abc or 0 were rejected. The asset list displayed all assets properly. Searching by name was successful whether it was entered as KENNEDY or kennedy (case ignored). Menu input of abc, enter or blank line, and 9 were rejected.

Reports

Having entered data in every module, the reports were consistent with the other modules. The employee report printed 104000 for an employee earning 100000 and 2000 houses and 2000 transport. The budget report printed 998000 as the remaining of a budget of 1000000 with 2000 spent and no departments over budget. The supplier report printed the registered suppliers, the asset report printed one asset with a total value of 140000, and the 'all reports' printed all four. Having not entered any data, every report printed a clear message to say nothing had been entered yet.

Checks not carried out

Several checks were not made in the time, they are: A blank Enter and 9999999999999 on the main menu, employee report with several employees on varying salaries, budget report with a department over budget, two budget departments with same name, search for suppliers by ID and Town, supplier name space only, supplier edit with an invalid phone number, 9 or x as asset type, and search for assets no match.